Final Report

FLEX LENS

A PROJECT REPORT

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Co-Supervisor Signature
(Ms. Bhawna Verma & E-11794)

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BONAFIDE CERTIFICATE

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HEAD OF THE DEPARTMENT

<<Department>>

Submitted for the project viva-voce examination held on 19.05.2022

INTERNAL EXAMINER **EXTERNAL EXAMINER**

ACKNOWLEDGEMENT

On this great occasion of accomplishment of our project FLEX LENS, we would like to sincerely express our gratitude to **Mr. Aniket Singh** and **Ms. Bhawna Verma** who have supported through the completion of this project. Whenever we faced any trouble, they were there for us.

Finally, as one of the team members, I would like to appreciate all my group members for their support and coordination. I hope we will achieve more in our future endeavors.

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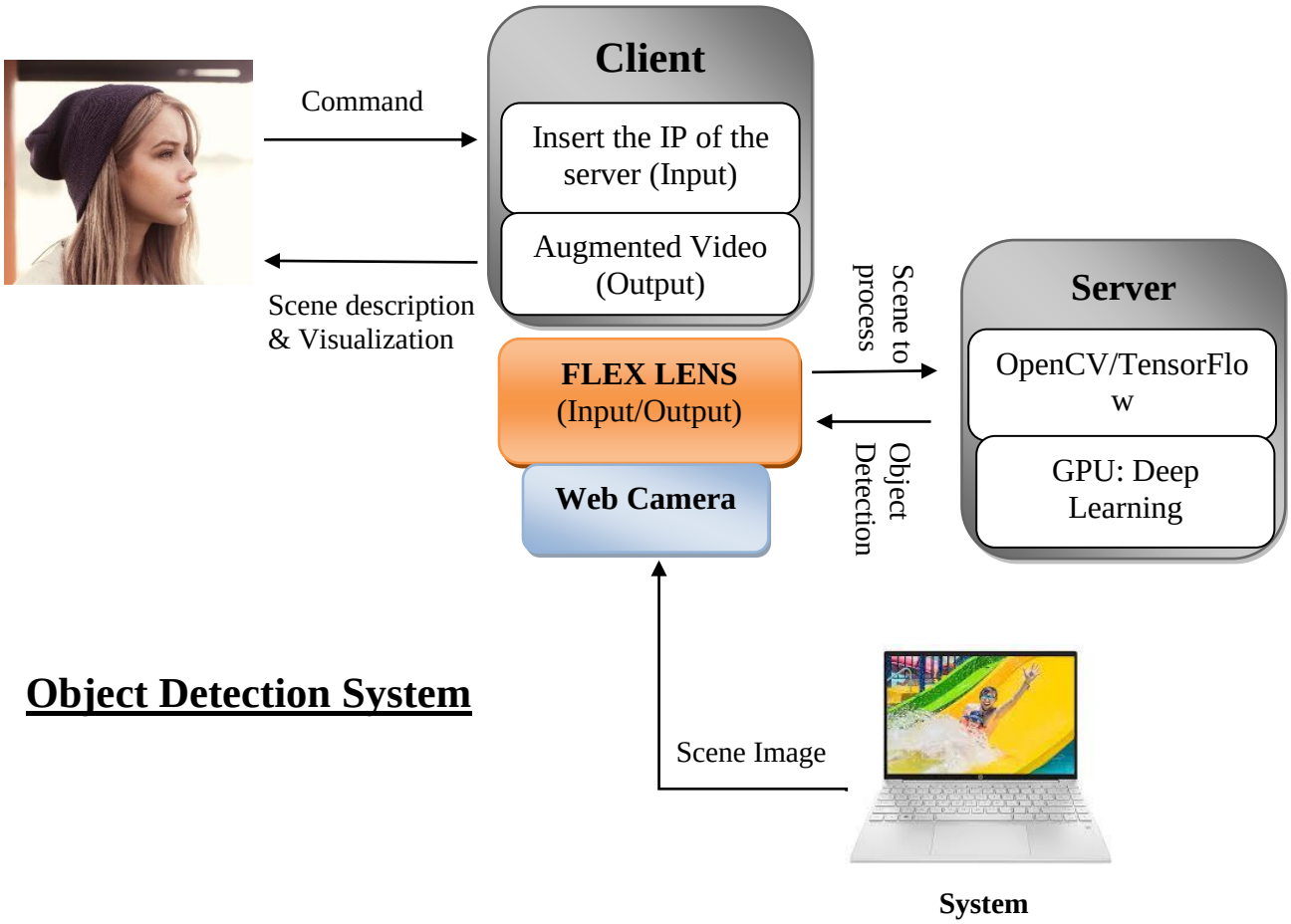
ABSTRACT

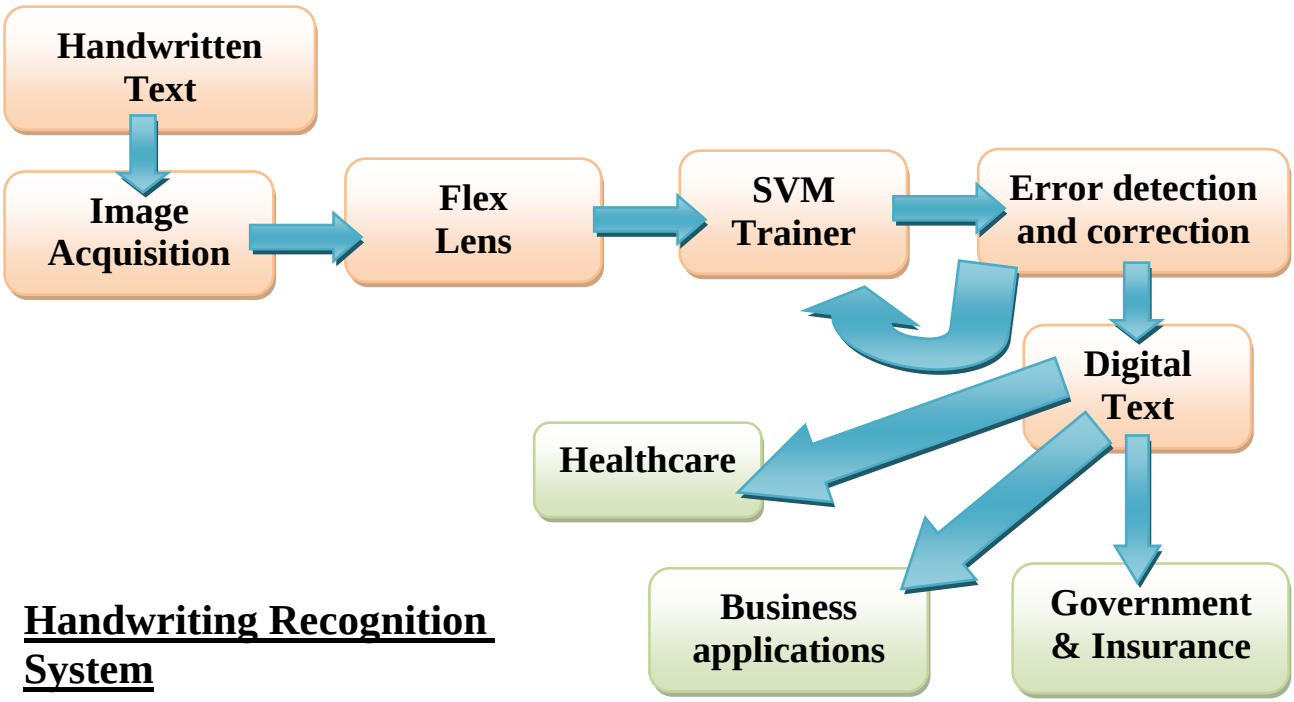
Object detection is one of the most basic and central tasks in computer vision. Its task is to find all the interested objects in the image, and determine the category and location of the objects. Object detection is widely used and has strong practical value and research prospects. Applications include face detection, pedestrian detection and vehicle detection. In recent years, with the development of convolutional neural network, significant breakthroughs have been made in object detection. This paper describes in detail the classification of object detection algorithms based on deep learning. The algorithms are mainly divided into one-stage object detection algorithm and two-stage object algorithm, and the general data sets and performance indicators of object indicators.

Handwritten character recognition (HCR) is the detection of characters from images, documents and other sources and changes them in machine-readable shape for further processing. The accurate recognition of intricate-shaped compound handwritten characters is still a great challenge. Recent advances in convolutional neural network (CNN) have made great progress in HCR by learning discriminatory characteristics from large amounts of raw data. In this paper, CNN is implemented to recognize the characters from a test dataset. The main focus of this work is to investigate CNN capability to recognize the characters from the image dataset and the accuracy of recognition with training and testing.

GRAPHICAL ABSTRACT



Object Detection System



Handwriting Recognition System

CHAPTER I

1.1 INTRODUCTION:

Computer Vision is the branch of the science of computers and software systems which can recognize as well as understand images and scenes. Computer Vision is consisting of various aspects such as image recognition, object detection, image generation, image super-resolution and many more. Object detection is widely used for face detection, vehicle detection, pedestrian counting, web images, security systems and self-driving cars. In this project, we are using highly accurate object detection-algorithms and methods such as R-CNN, Fast-RCNN, Faster-RCNN, RetinaNet and fast yet highly accurate ones like SSD and YOLO. Using these methods and algorithms, based on deep learning which is also based on machine learning require lots of mathematical and deep learning frameworks understanding by using dependencies such as TensorFlow, OpenCV, imageAi etc., we can detect each and every object in image by the area object in a highlighted rectangular box and identify each and every object and assign its tag to the object. This also includes the accuracy of each method for identifying objects

Handwriting digits and character recognitions have become increasingly important in today's digitized world due to their practical applications in various day to day activities. It can be proven by the fact that in recent years, different recognition systems have been developed or proposed to be used in different fields where high classification efficiency is needed. Systems that are used to recognize Handwriting letters, characters, and digits help people to solve more complex tasks that otherwise would be time-consuming and costly. A good example is the use of automatic processing systems used in banks to process bank cheques. Without automated bank cheque processing systems, the bank would be required to employ many employees who may not be as efficient as the computerized processing system. The handwriting recognition problem can be considered for various alphabets and at various levels of abstraction. The main goal of the work presented in this paper has been the development of an on-line handwriting recognition system which is able to recognize handwritten characters of several different writing styles. This approach has succeeded in having robust pattern recognition features, while maintaining feature's domain space to a small, optimum quantity. Back propagation neural network (BPN) technique has been used as classifier and recognition rate up to 87% has been achieved even for highly distorted hand written characters.

1.2 PROBLEM DEFINITION:

Iterating over the problem of localization plus classification we end up with the need for detecting and classifying multiple objects at the same time. Object detection is the problem of finding and classifying a variable number of objects on an image. The important difference is the "variable" part. In contrast with problems like classification, the output of object detection is variable in length, since the number of objects detected may change from image to image. In this post we'll go into the details of practical applications, what are the main issues of object detection as a machine learning problem and how the way to tackle it has been shifting in the last years with deep learning.

Handwriting character recognition is one of the research fields in computer vision, artificial intelligence, and pattern recognition. A computer application that performs handwriting recognition can be argued to have the ability to acquire and detecting characters in pictures, paper documents, and other sources and convert them into electronic format or machine-encoded form. The system may obtain Handwriting sources from a piece of paper through optical scanning or intelligent word recognition. Also, the system may be designed to detect the movement of the pen tip on the screen. In other words, handwriting recognition may involve a system detecting movements of a pen tip on the screen to get a clue of the characters being written.

Handwriting recognition can be classified into two:

- **Offline recognition**

Offline handwriting recognition involved the extraction of text or characters from an image to have letter codes that can be used within a computer. It involves obtaining digital data from a static representation of handwriting. A system is provided with a Handwriting document to read and convert the handwriting to a digital format.

- **Online recognition**

Online handwriting recognition, on the other hand, involved automatic detection or conversion of characters as they are written on the specialized screen. In this case, the system sensors movement of pen-tip to detect characters and words. Different methods and techniques are used to ensure that computer systems can read characters from Handwriting images and documents.

Among the existing techniques that are used to model, and train Handwriting character recognition include neural network, Hidden Markov Model (HMM), Machine Learning, and Support Vector Machine, to mention a few. This paper focuses on artificial intelligence networks, machine learning, Hidden Markov Model, and the Support Vector Machine.

CHAPTER II

LITERATURE SURVEY

Deep convolutional neural networks recently demonstrated very impressive performance on a number of image recognition benchmarks. It has shown good performance on large scale visual recognition challenge. It was a winning model on localization subtask with the process by predicting single bounding box and identifying object category in the image. But the model cannot handle multiple instances of same object in the image. But the model cannot handle multiple instances of same object in the image. But now it can handle the same image having multiple instances and allows cross class generalization at highest level of network. In this paper the computational challenge is addressed. Also, this challenge becomes even harder when an object occurs more than once in the image. How they tackle this by generating a no of bounding boxes. For each box the output is a confidence score i.e., the likelihood of an image existing in that box. Various training exercises are performed for this. The predicted results and the real results are then matched for the learning purpose. They use the Deep Neural Network which produces a fixed number of bounding boxes and then gives the output of each box as a confidence score.

Recurrent neural network (RNN) based language model (RNNLM) is a biologically inspired model for natural language processing. It records the historical information through additional recurrent connections and therefore is very effective in capturing semantics of sentences. At architectural level the parallelism of RNN training scheme is improved and also reduces the computing resource requirement. Experiments at different network sizes demonstrates a great scalability of proposed framework. RNN is a different type of neural network that can operate in time domain. RNN captures the long-range dependencies using the additional recurrent connection. Then it stores them in hidden layer for later use. But the training costs in RNN was really high. So hardware up gradation was necessary to make it feasible. FPGA based accelerators have really caught the attention for tackling this problem. Modern language models are based on statistical analysis. Thus, a proper balance between computation unit and memory bandwidth should be obtained be obtained by proper scalability. Next comes the architectural optimization. It has various things to do in it. Like to increase parallelism between output layers and hidden layers. But it can only be done to a certain extent as there are limitations to it. Then

there are the hardware implementations. The FPGA hardware design plays a huge role in supporting RNN

Online handwriting recognition online handwriting recognition, where existing challenges are to cope with problems of various writing fashions, variable size for the same character, different stroke orders for the same letter, and efficient data presentation to the classifier. The similarities of distinct character shapes and the ambiguous writing further complicate the dilemma. A solitary solution of all these problems lies in the intelligent and appropriate extraction of features from the character at the time of writing. A typical handwriting recognition system focuses on only a subset of these problems.

CHAPTER III

3.1 CHARACTERISTIC IDENTIFICATION:

Objective

- The main purpose of object detection is to identify and locate one or more effective targets from still image or video data. It comprehensively includes a variety of important techniques, such as image processing, pattern recognition, artificial intelligence and machine learning.
- It has broad application prospects in such areas such as road traffic accident prevention, warnings of dangerous goods in factories, military restricted area monitoring and advanced human–computer interaction.
- Since the application scenarios of multi-target detection in the real world are usually complex and variable, balancing the relationship between accuracy and computing costs is a difficult task

Single Entity

- Execution of the undertaking with a little group is a quite extreme errand, yet we figured out how to keep the work underway by partitioning the venture in modules and teaming up and guaranteeing the task is finished as a Single Entity

Lifespan

- The project is divided into two parts i.e., “Object detection system” as the 1st part and “Handwriting recognition system” as the 2nd part. Hence, time period of 1 month id required for completion of the process

Require Funds

- The project is a combined product of hardware and programmed interface. The cost depends on the number of cameras and the miscellaneous circuitry.

  Thus, the setup cost of 1 camera set is 10,000/- INR.

Life Cycle

- The undertaking lifecycle began with formulating out the thought for the task.

- After effectively concluding the task, our group began to design out how the beginning stage will be executed.
- The significant terms in regards to the task were characterized and we began with the plan interaction.
- Subsequent to finishing the planning of the undertaking, the task was worked in around multi month time, being tried for any issues and afterward it will be sent as the last advance of the improvement lifecycle.

Team Spirit

- All through the improvement interaction, our group crossed numerous snags to accomplish the ideal outcomes. At times it took us days to determine a single bug which was very baffling as the advancement ended. In any case, keeping our expectations high and keeping us loaded up with enthusiasm and inspiration, we continued pushing ahead beating every one of the obstructions and effectively finished the undertaking

Risk and Uncertainty

- Project advancement represents a ton of hazard and vulnerabilities which should be dealt with during the improvement interaction. Some of them include:
    - Virtual environment crash while setting it up using Jupyter notebook in Anaconda Navigator.
    - Image and video lagging during testing period leading lead to creation of new start from the scratch to detect bugs.
    - Hardware and software incompatibility due to version differences.
    - Financial losses due to purchase of different sets of cameras for software compatibility and efficient functioning tests.
- Keeping a note of this large number of dangers, we worked interminably to adapt to this multitude of vulnerabilities and thought of an answer which limits the above angles.

Directions

- In the wake of fostering the model of the setup, we took client input about what the machine needs and where it required a few alterations.

- In view of the inputs got, we worked in the headings given by them to construct a machine that welcomes clients.

Uniqueness

- The algorithm used in our machine setup and the component compatibility gives our machine an edge and defines its uniqueness and further workforce for a better, smarter and efficient machine setup will lead us to patent acquisition with ease.
- Our team has trained the AI with numerous test subjects leading to 4-digit testing series.
- With upgradation in the interface and alteration won’t lead compromise in its functionality, because we have assigned two members of our team for its testing.

Flexibility

- The task has been planned so that it is generally open to changes as expected by the client. The task is adaptable as far as utilization and changes can be made progressively whenever required

Sub-contracting

- Our group has complete information about each progression of the undertaking improvement cycle and needed exceptionally less external support. However, we took help from our co-supervisor who directed us to make this venture a triumph.

Cost

- As referenced about the expense expected for the undertaking of 1 camera setup, regardless of whether the venture goes through certain changes, the cost won't be affected a lot.
- Need of number of camera setup is based on the field of usage.

3.2 CONSTRAINTS IDENTIFICATION:

Time

- During the arranging period of the advancement lifecycle, we zeroed in on how long we would expect for project completion.
- This guaranteed that we worked inside the time constraints.

Cost

- While sorting out the funds expected to effectively complete the venture, we put forth a most extreme line we can spend for the undertaking and guaranteed that regardless of whether additional assets are required, the cost constraints won't be impacted.

Scope

- As currently referenced in the venture plan, our undertaking is fairly interesting and many haven't planned anything like this. Therefore, the scope constraints preclude from the conversation as we did a very great job by the decision of the venture

Quality

- Our group has made an honest effort to hold the nature of the venture within proper limits. Rest, the client input will permit us to make alterations as required monitoring the quality constraints

Benefits

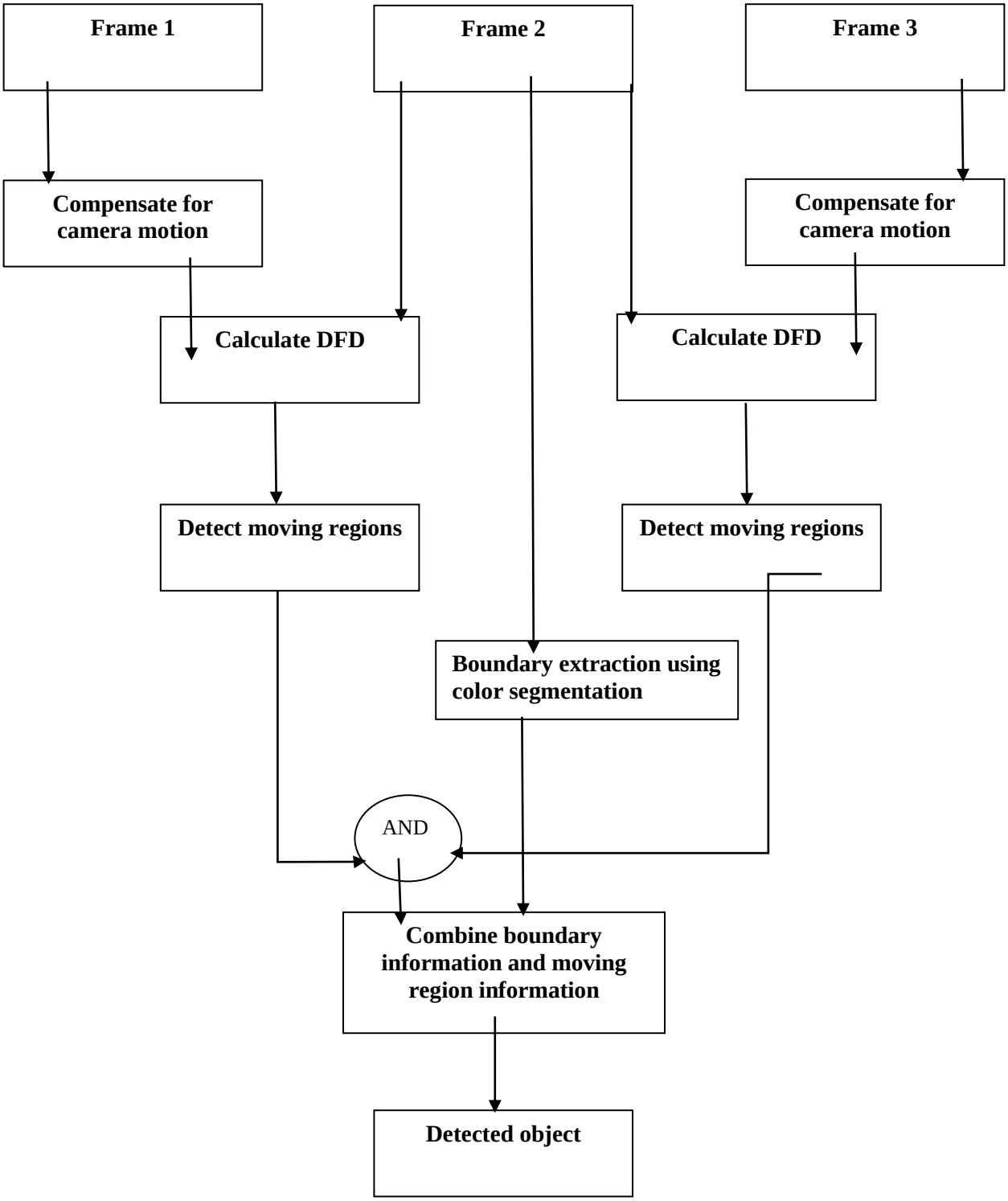
- Being one of a kind in its own particular manner, it has a ton of advantages. The advantages of the application don't appear to stop just like a stage which is connected with endlessly learning is a continuous cycle in any individual's life

Risk

- As currently talked about, the improvement of an undertaking offers a great deal of dangers that might be of some value, however holding them within proper limits is what we truly need to focus on.

3.3 DESIGN FLOW DIAGRAMS:

DFD Model 1:



DFD Model 2:

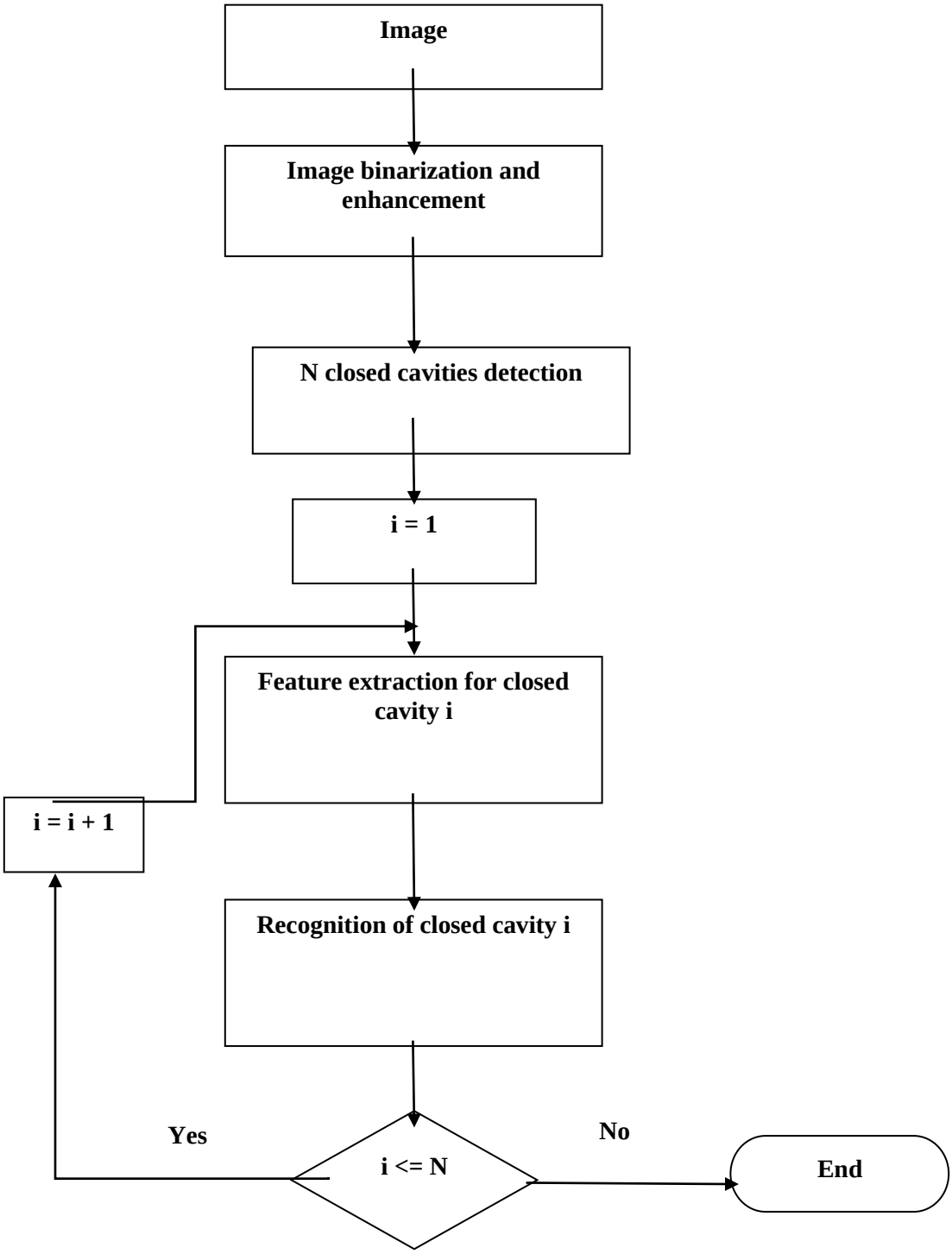
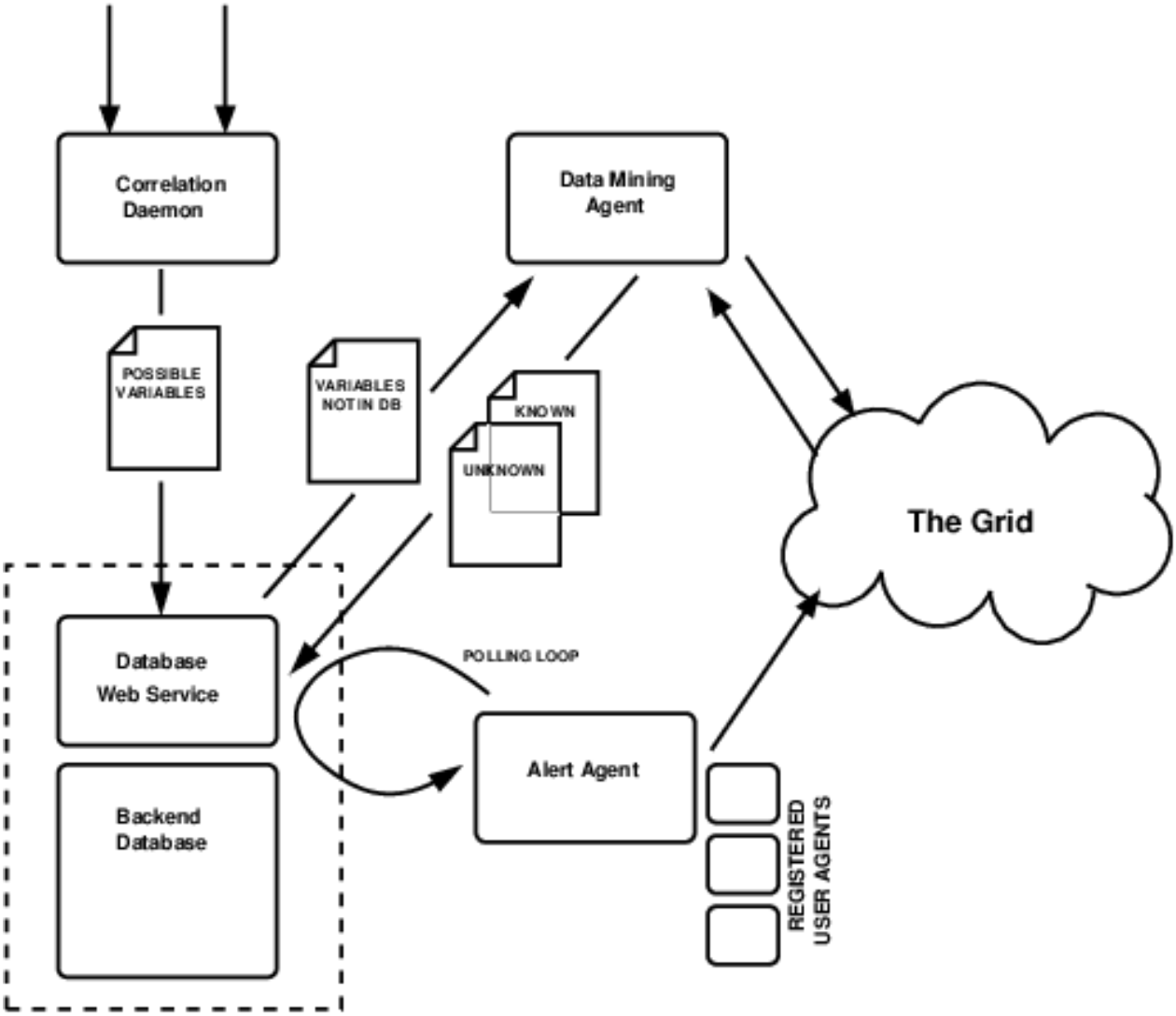
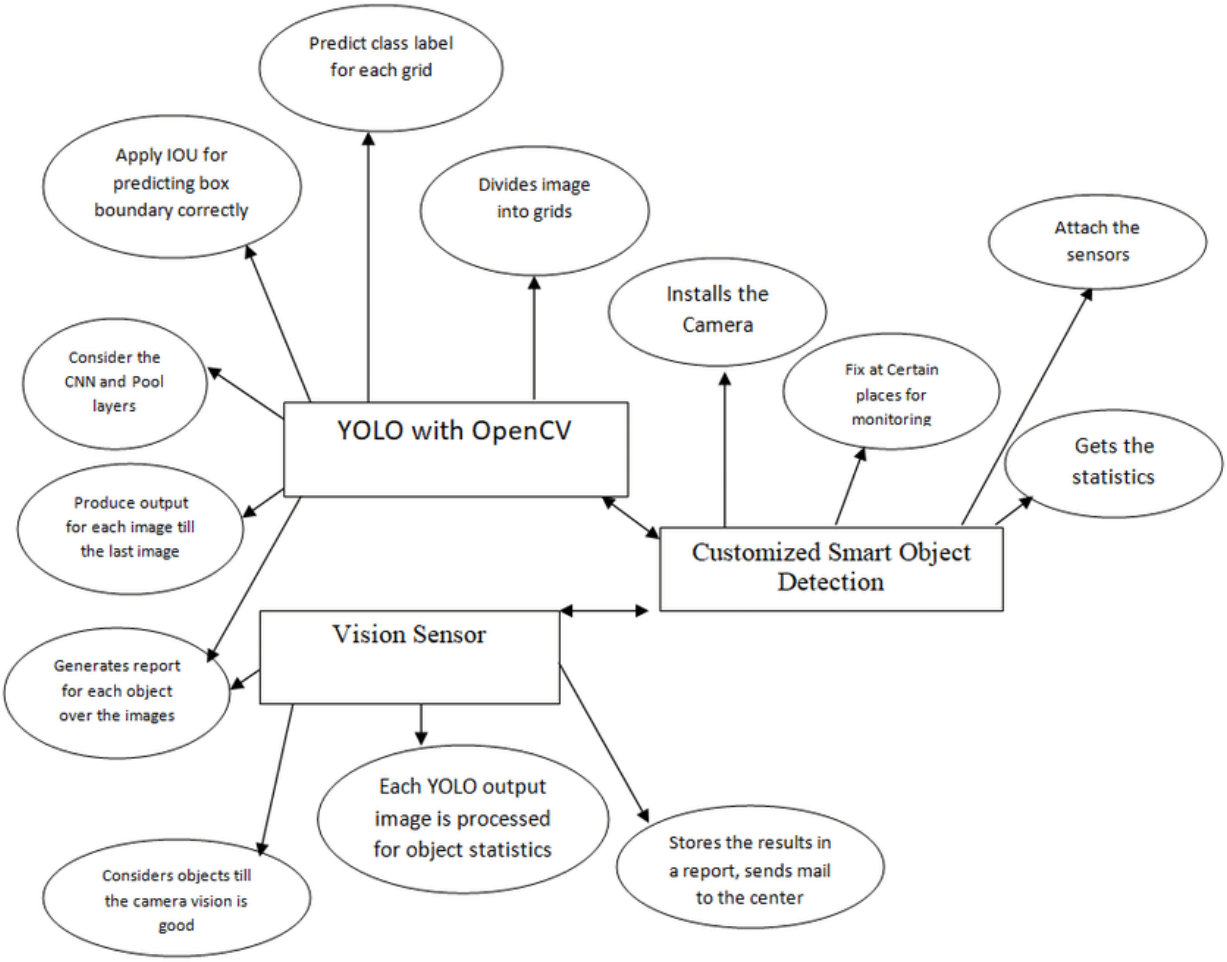


Image
Image binarization and enhancement
N closed cavities detection
i = 1
Feature extraction for closed cavity i
i = i + 1
Recognition of closed cavity i
Yes
i <= N
No
End

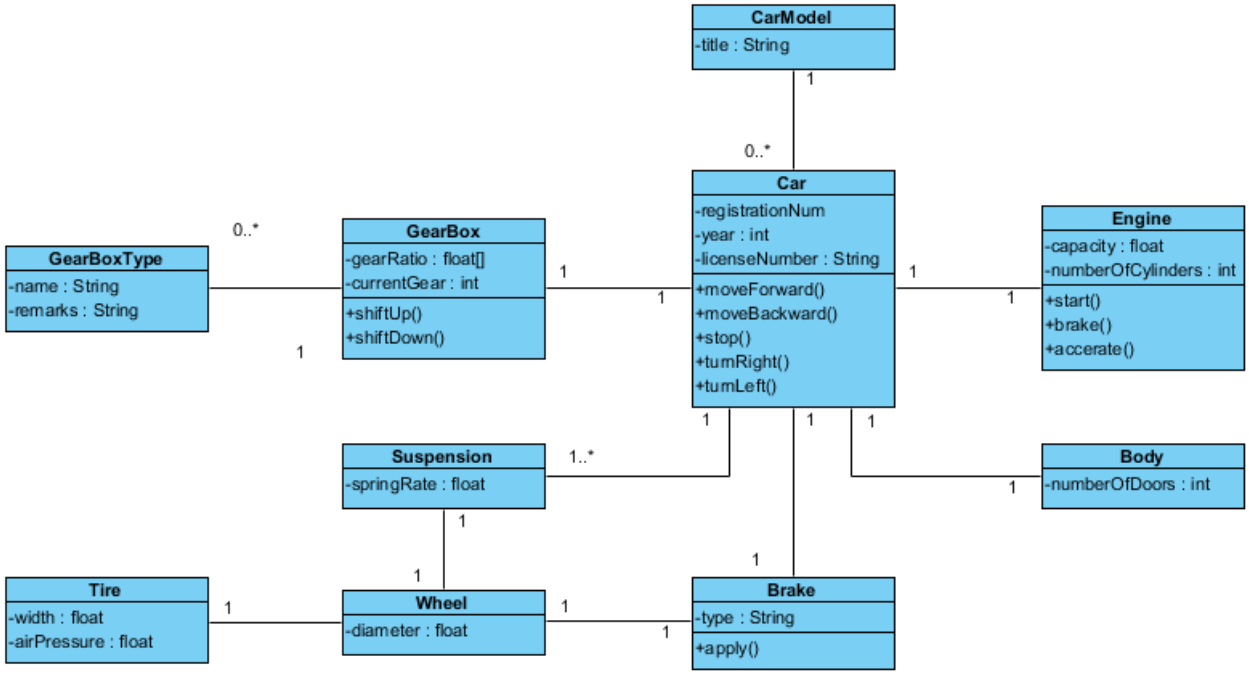
DFD Model 3:


Correlation Daemon
Data Mining Agent
POSSIBLE VARIABLES
VARIABLES NOT IN DB
KNOWN
UNKNOWN
The Grid
Database Web Service
POLLING LOOP
Alert Agent
Backend Database
REGISTERED USER AGENTS

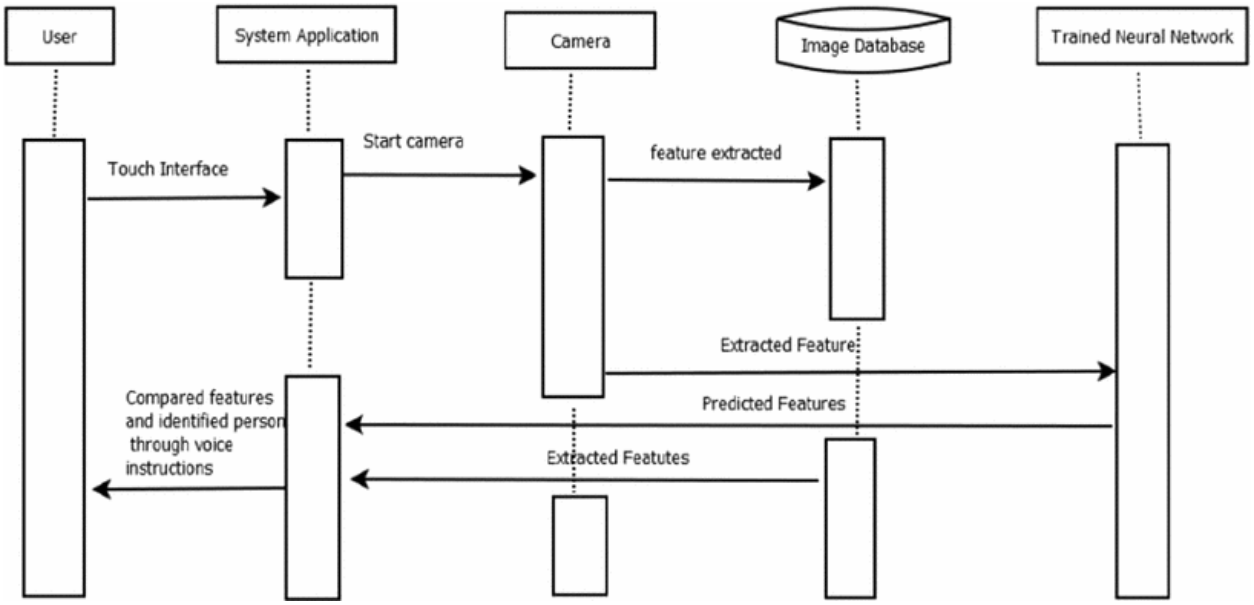
ER- Diagram:


Predict class label for each grid
Apply IOU for predicting box boundary correctly
Divides image into grids
Attach the sensors
Installs the Camera
Consider the CNN and Pool layers
Fix at Certain places for monitoring
YOLO with OpenCV
Gets the statistics
Produce output for each image till the last image
Customized Smart Object Detection
Vision Sensor
Generates report for each object over the images
Each YOLO output image is processed for object statistics
Stores the results in a report, sends mail to the center
Considers objects till the camera vision is good

Class Diagram:



Sequence Diagram:



CHAPTER IV

4.1 Various Technology used:

Computer Vision is the branch of the science of computers and software systems which can recognize as well as understand images and scenes. Computer Vision is consisting of various aspects such as image recognition, object detection, image generation, image super-resolution and many more. Object detection is widely used for face detection, vehicle detection, pedestrian counting, web images, security systems and self-driving cars.

Handwriting digits and character recognitions have become increasingly important in today's digitized world due to their practical applications in various day to day activities. It can be proven by the fact that in recent years, different recognition systems have been developed or proposed to be used in different fields where high classification efficiency is needed. Systems that are used to recognize Handwriting letters, characters, and digits help people to solve more complex tasks that otherwise would be time-consuming and costly.

Software and Libraries Requirement:

- Python 3.7.2
- Anaconda navigator
- Jupyter Notebook
- Tensorflow
- Matplotlib
- Numpy

1. **Anaconda navigator:**

   Anaconda Navigator is a desktop graphical user interface (GUI) included in Anaconda® distribution that allows you to launch applications and easily manage conda packages, environments, and channels without using command-line commands. Navigator can search for packages on Anaconda.org or in a local Anaconda Repository.

2. **Jupyter Notebook:**

The Jupyter Notebook is the original web application for creating and sharing computational documents. It offers a simple, streamlined, document-centric experience.

3. **Tensorflow:**

TensorFlow is a free and open-source software library for machine learning and artificial intelligence. It can be used across a range of tasks but has a particular focus on training and inference of deep neural networks.

4. **Matplotlib:**

Matplotlib is a Python programming language plotting library and its NumPy numerical math extension. It provides an object-oriented API to use general-purpose GUI toolkits suchasTkinter, wxPython, Qt, or GTK+ to embed plots into applications.

pip install matplotlib – command

5. **Numpy:**

NumPy is library of Python programming language, adding support for large, multi-dimensional array and matrices, along with large collection of high-level mathematical function to operate over these arrays. The ancestor of NumPy, Numeric, was originally created by Jim Hugunin with contributions from several developers. In 2005 Travis Olphant created NumPy by incorporating features of computing Num array into Numeric, with extension modifications. NumPy is open-source software and has many contributors.

pip install numpy -command

CHAPTER V

5.1 CONCLUSION:

Deep-learning based object detection has been a search hotspot in recent years. This project starts on generic object detection pipelines which give base architectures for other related tasks. With the assistance of this the 3 other common tasks, namely object detection, face detection and pedestrian detection, are often accomplished. Authors accomplished this by combing 2 things: Object detection with deep learning and OpenCV and Efficient, threaded video streams with OpenCV. The camera sensor noise and lightening condition can change the result because it can create problem in recognizing the objects. generally, this whole process requires GPU's rather than CPU's. But we've done using CPU's and executes in much less time, making it efficient. Object Detection algorithms act as a mixture of both image classification and object localization. It takes the given image as input and produces the output having the bounding boxes adequate to the number of objects present within the image with the category label attached to every bounding box at the highest. It projects the scenario of the bounding box up the shape of position, height and width.

5.2 FUTURE SCOPE:

The proposed feature extraction method helps to select the important feature. To improve the efficiency of the classifier, the number of features should be less in number. Specifically, the contributions towards this research work are as follows,

- An object recognition system is developed, that recognizes the two-dimensional and three-dimensional objects.
- The feature extracted is sufficient for recognizing the object and marking the location of the object. x the proposed classifier is able to recognize the object in less computational cost.
- The proposed global feature extraction requires less time, compared to the traditional feature extraction method.
- The performance of the SVM-kNN is greater and promising when compared with the BPN and SVM.
- The performance of the One-against-One classifier is efficient

<u>As a scope for future enhancement,</u>

- Features either the local or global used for recognition can be increased, to increase the efficiency of the object recognition system.
- Geometric properties of the image can be included in the feature vector for recognition. 150
- Using unsupervised classifier instead of a supervised classifier for recognition of the object.
- The proposed object recognition system uses grey-scale image and discards the color information.

  The color information in the image can be used for recognition of the object. Color based object recognition plays vital role in Robotics

Although the visual tracking algorithm proposed here is robust in many of the conditions, it can be made more robust by eliminating some of the limitations as listed below:

- In the Single Visual tracking, the size of the template remains fixed for tracking. If the size of the object reduces with the time; the background becomes more dominant than the object being tracked. In this case the object may not be tracked.
- Fully occluded object cannot be tracked and considered as a new object in the next frame.
- Foreground object extraction depends on the binary segmentation which is carried out by applying threshold techniques. So, blob extraction and tracking depend on the threshold value.
- Splitting and merging cannot be handled very well in all conditions using the single camera due to the loss of information of a 3D object projection in 2D images
- For Night time visual tracking, night vision mode should be available as an inbuilt feature in the CCTV camera.

  To make the system fully automatic and also to overcome the above limitations, in future, Multiview tracking can be implemented using multiple cameras. Multi view tracking has the obvious advantage over single view tracking because of wide coverage range with different viewing angles for the objects to be tracked.

5.3 USER MANUAL:

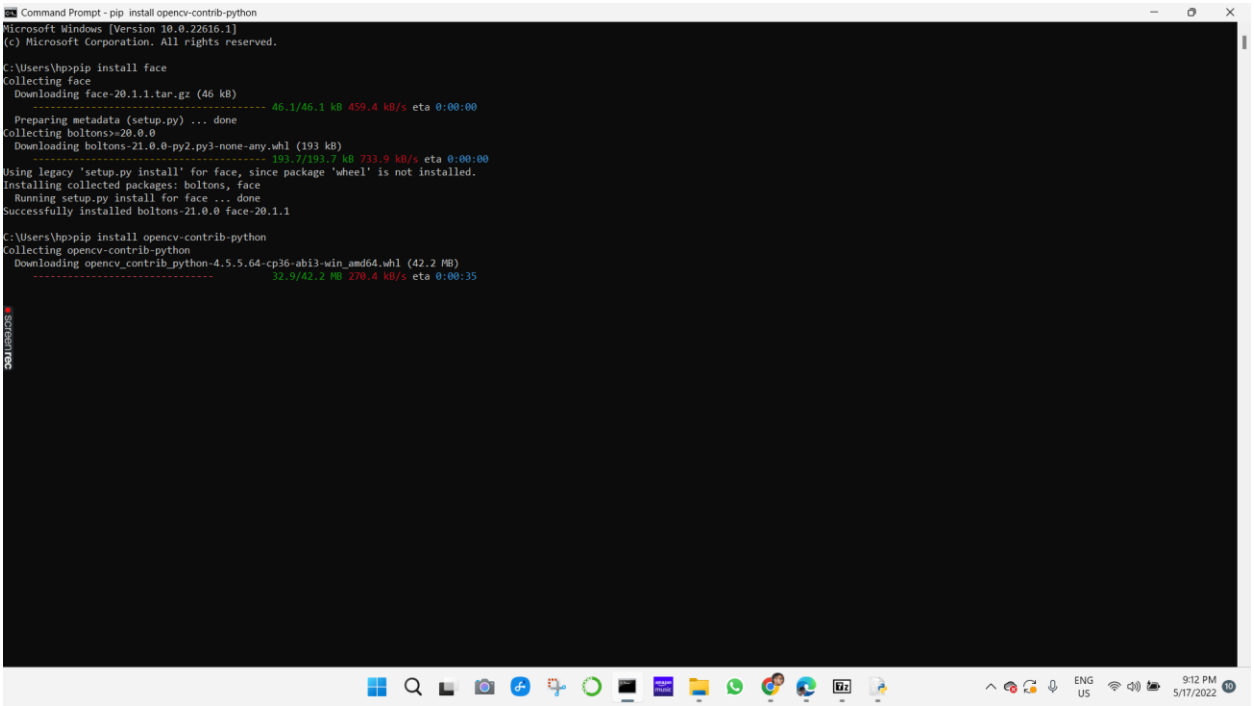
1. Step wise guide for object detection system

<u>Step-1</u>

- Open the command prompt and type the below statements and download the packages.

$pip install face

$opencv-contrib-python

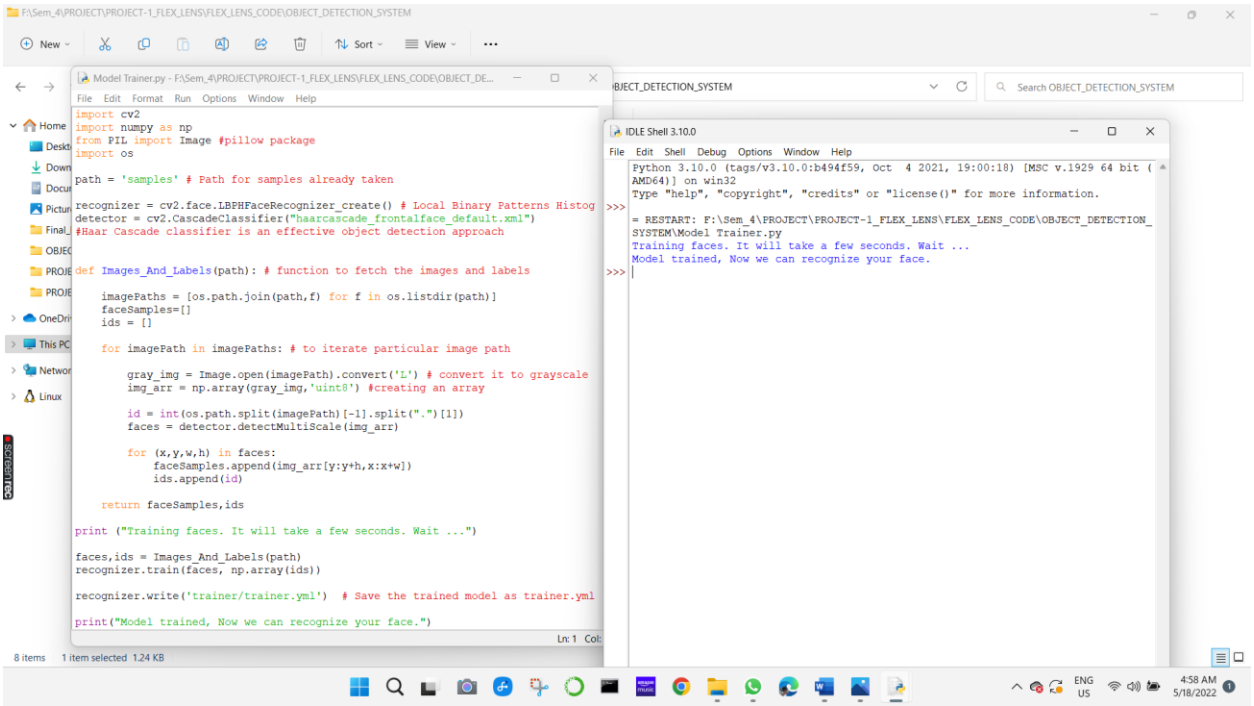


<u>Step-2</u>

- Open the file saved as “Sample Generator.py”
- Click on “Run”
- Click on “Run module”
- Pic samples gets captured and stored in the folder named as “samples”

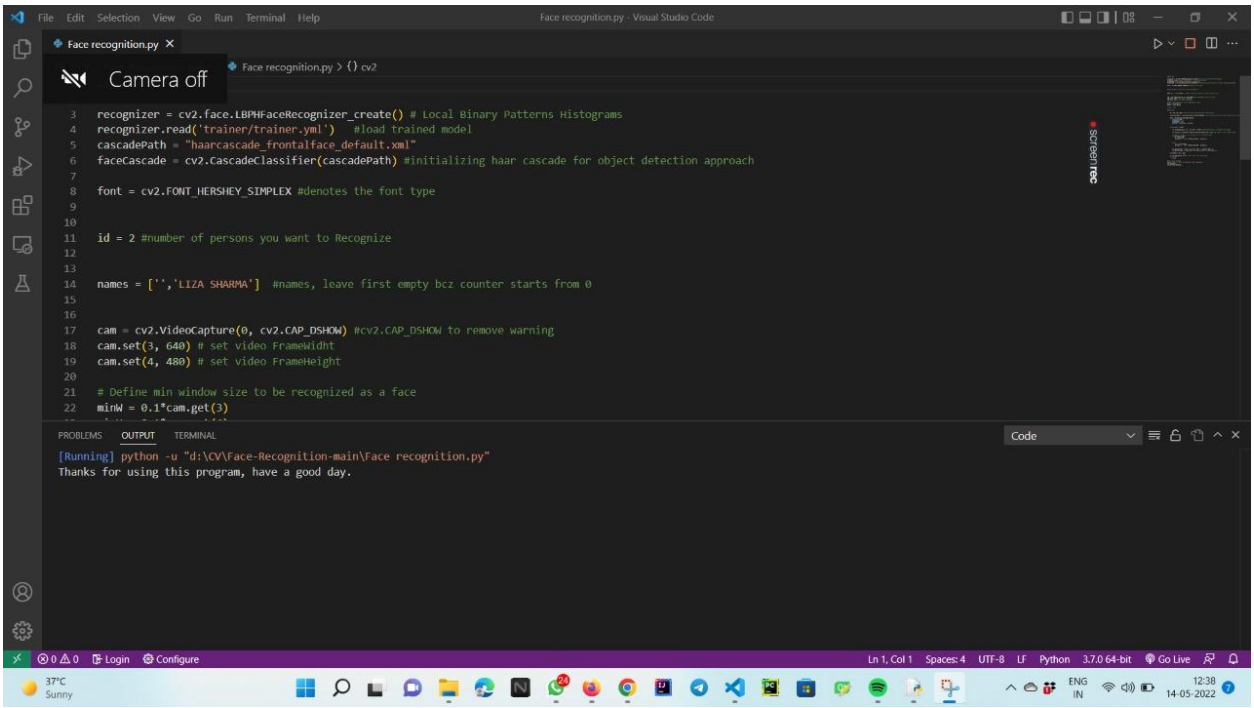
Step-3

- Open the file saved as “Model Trainer.py”
- Click on “Run”
- Click on “Run module”



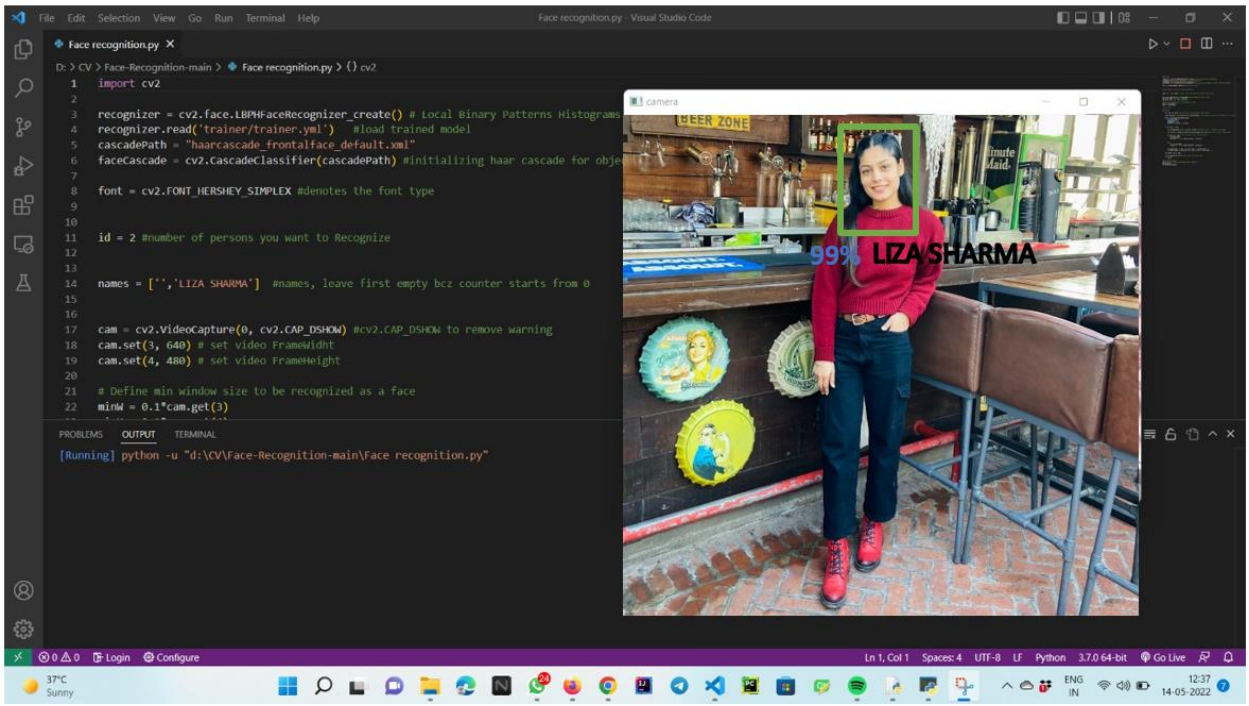
Step-4

- Open the file saved as “Face recognition.py”
- The dialog box with code appears.
- Enable the camera and press run.



Step-5

- Human face detected.



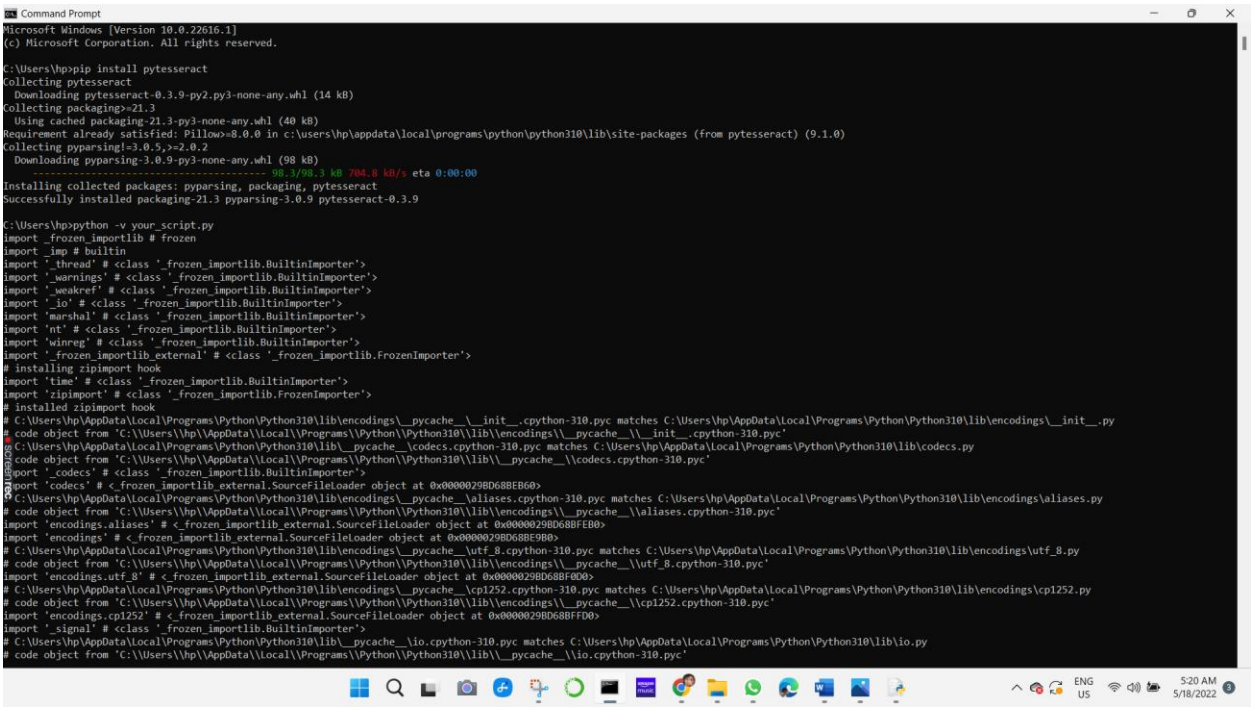
2. Step wise guide for handwriting recognition system

<u>Step-1</u>

- Open the command prompt and type the below statements and download the packages.

$pip install pytesseract

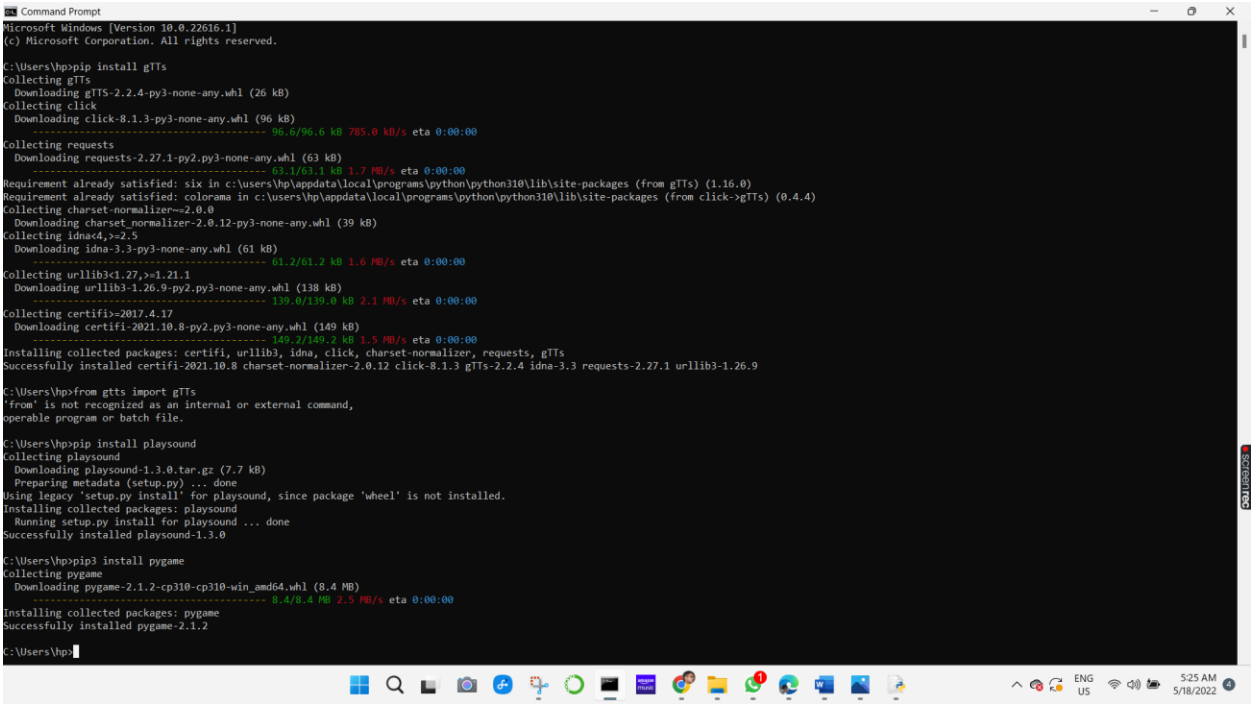
python -v your_script.py



pip install gTTs

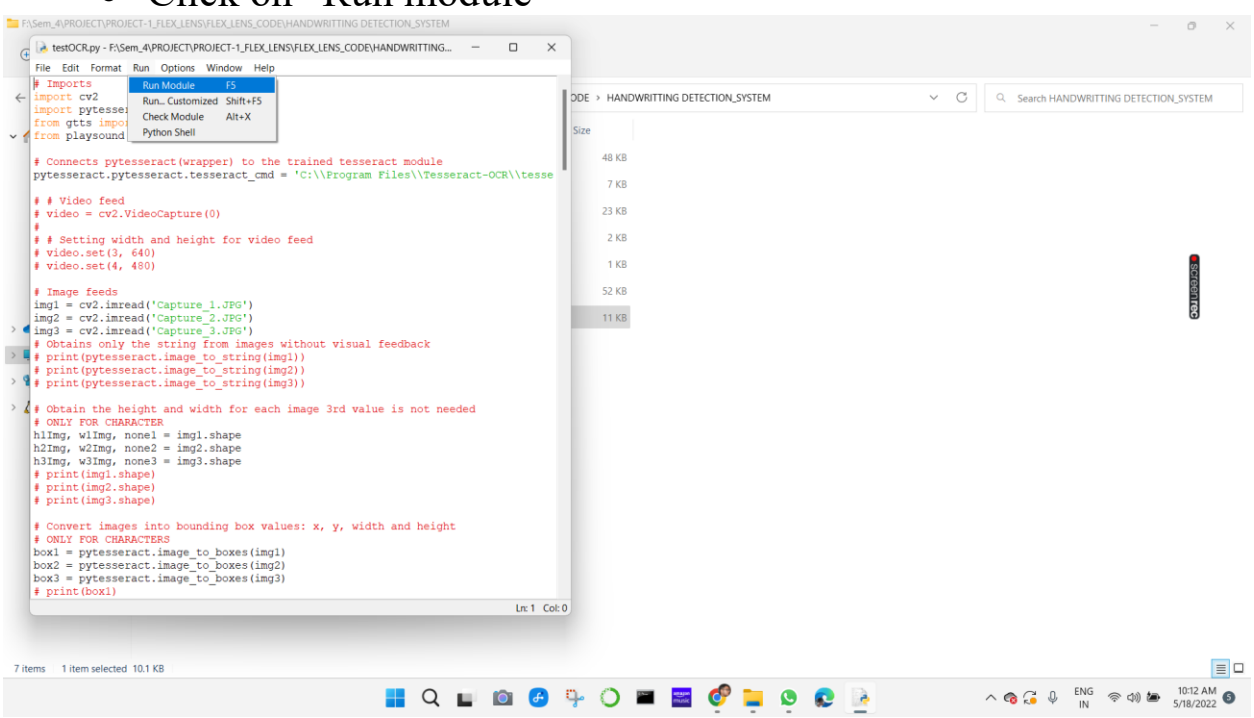
pip install playsound

pip3 install pygame



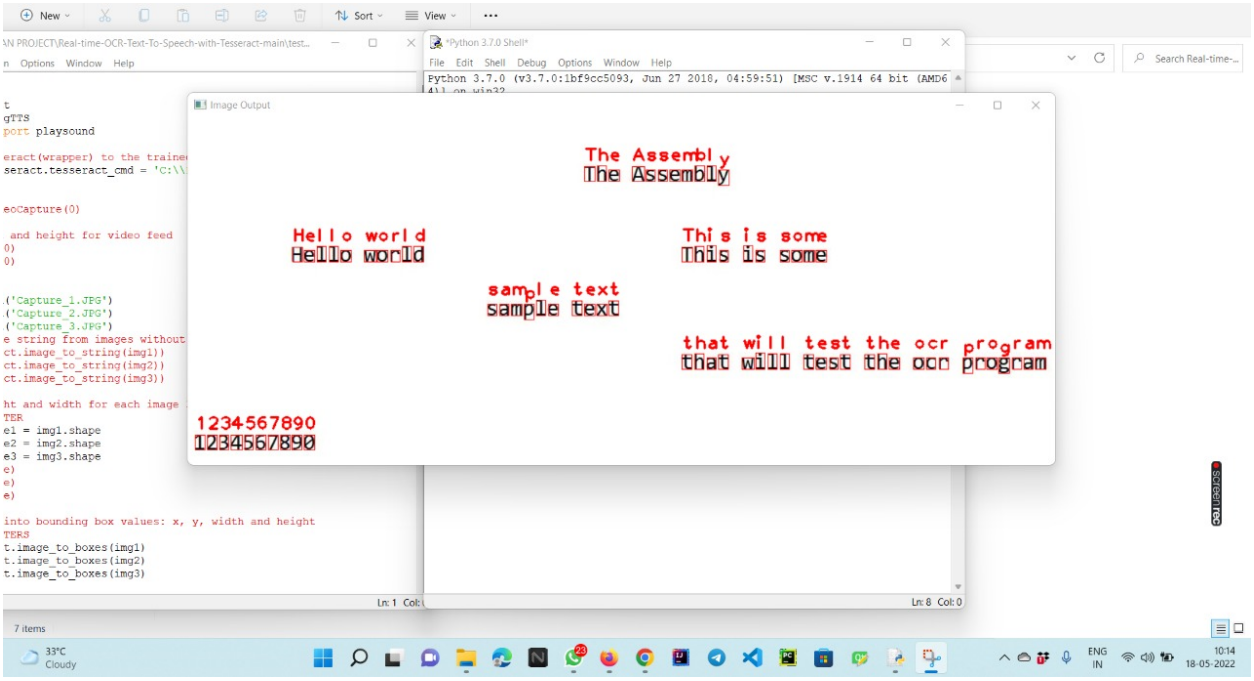
Step-2

- Open the file saved as “Model Trainer.py”
- Click on “Run”
- Click on “Run module”



STEP-3

- HANDWRITTEN TEXT RECOGNIZED BY SYSTEM AND PROJECT EXECUTED SUCCESSFULLY.



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